Simple Linear Regression

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Chapter overview

This chapter reads regression as prediction with quantified uncertainty. It opens with the simple regression model, a straight line for the mean of an outcome with three parameters, and reads those parameters in a picture and in a simulated process whose values we know. It then turns to a reported claim about education and income and a question about 102 occupations: we plot the records, fit a straight line, read its slope, intercept, fitted values and residuals as a description of those records, and see how least squares chooses the line and how well it fits. Declaring the model for the occupations gives the slope a standard error and an interval, repeated samples of the known process show what the interval's coverage means, and prediction comes at three levels: a point, a mean response and one new case. Checks of form, spread, influence and support qualify these answers. The chapter closes with regression toward the mean in its two versions, Galton's heights of fathers and sons and a hypothetical class's two exams. By the end you will be able to state and read a simple regression model, fit and read one in R, say what its descriptive and model-based answers concern, and recognize regression toward the mean in a before-and-after claim. An appendix develops the mathematics.

Suppose a report says, "Another year of education increases income by about 900 dollars." Before asking whether the number is significant, ask whose education and whose income were measured: individuals, households and occupations give three different comparisons.

Our question is narrower:

> Among the 102 recorded Canadian occupations, how did occupation-average income differ across the supported range of occupation-average education in 1971, as summarized by a fitted straight line?

One row is an **occupation**, not a worker. An occupation with higher average education need not contain the same people as one with lower average education, so a comparison of their averages does not describe what another year of schooling would do to one person's income. Aggregate and individual relationships can differ substantially (Robinson, 1950).

Regression can be read as **prediction with quantified uncertainty**. A fitted line predicts an occupation's average income from its average education; the residuals show how far each record lies from its prediction; and, once a model for the records is declared, standard errors and intervals say how uncertain the slope and the predictions are. Section 1 states the model that gives a slope its meaning; Section 2 returns to the occupations.

1 The Simple Regression Model

1.1 From a mean to a line

One common mean describes the centre of an outcome for all cases. Allowing separate means for groups recognizes that different categories may have different averages. Simple regression takes another step: the mean changes continuously along a straight line as a numerical predictor changes. The **conditional mean** is the mean outcome at a stated predictor value.

For case i, write x_i for its recorded predictor value and Y_i for an outcome the model could generate. We hold the complete set of observed predictor values $x_1, \ldots, x_n$ fixed. Repetition means new outcomes at those same values, so expectations and variances below suppress conditioning on that fixed set. The mean and individual outcome are

$$E(Y_i) = \beta_0 + \beta_1 x_i, \qquad Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The error ε_i is how far case i's outcome lies above or below the true mean line. The conditions

$$E(\varepsilon_i) = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2$$

say that errors average zero at every recorded predictor value and have the same variance everywhere. Two more conditions complete the model used for exact inference. The errors are independent of one another, a property of how cases were sampled, assigned or measured; a good-looking plot cannot establish it. For the exact t procedure of Section 3 they are also Normal, $\varepsilon_i \overset{\text{ind}}{\sim} \mathcal{N}(0, \sigma^2)$. Neither the distribution of the predictor values nor the pooled outcomes needs to be Normal: the condition concerns errors around the line at each predictor value. The predictor values must differ, since points that all share one x cannot determine a slope.

The model has three parameters: the **intercept** β_0, the **slope** β_1 and the **error SD** σ. They belong to the process that produces the data, and in practice they are unknown. A hat marks a quantity computed from data: $\widehat{\beta}_0$ and $\widehat{\beta}_1$ estimate β_0 and β_1, and the residual SD s_e of Section 2.6 estimates σ.

1.2 Reading the parameters

Regression is used to predict, and its parameters are read as comparisons. Figure 1 draws each part of the model on six invented points around the line $y = 2 + 0.5x$.

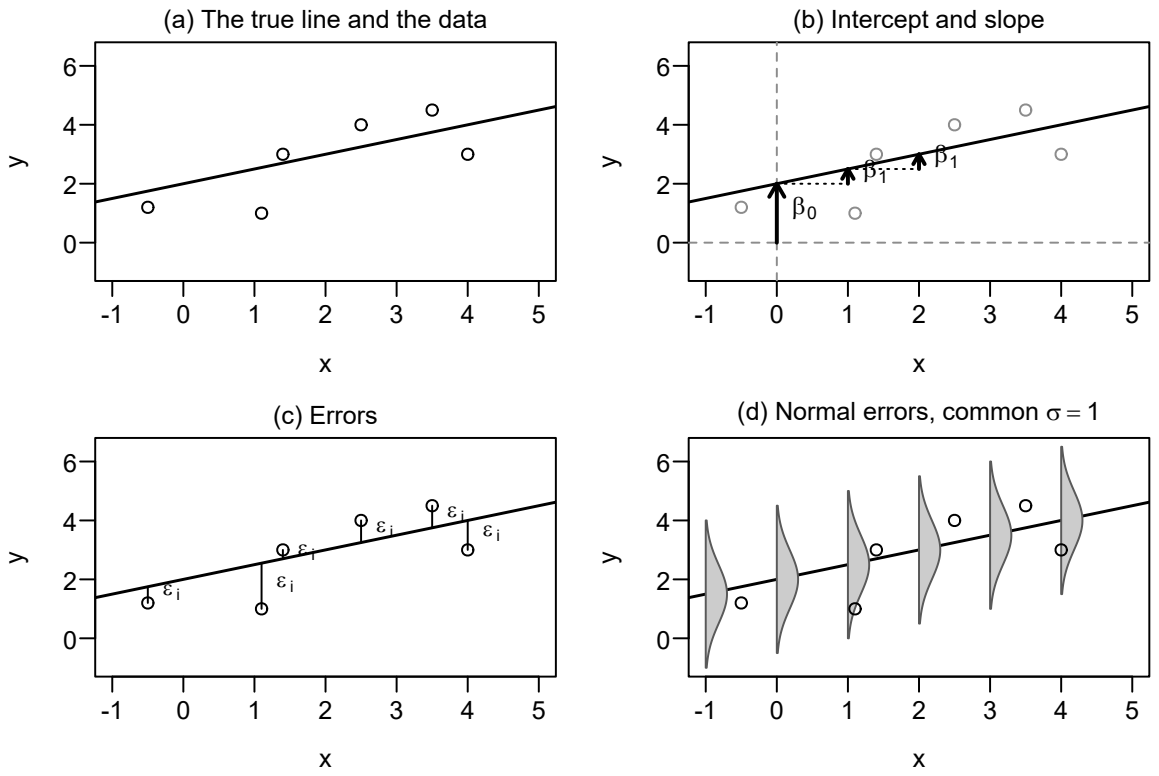


Figure 1: The parts of the simple regression model, drawn on six invented points around the true line $y = 2 + 0.5x$. (a) The true line and the data. (b) The intercept β_0 is the mean of y at $x = 0$, and the slope β_1 is the difference in mean y for a one-unit difference in x. (c) The errors ε_i are the vertical distances from the points to the true line. (d) The errors follow one Normal distribution around the line, with the same SD, here $\sigma = 1$, at every x.

Panel (a) shows the true line, the mean of y at each x, with six observed points around it. In panel (b) the slope, $\beta_1 = 0.5$, is the difference in mean y between cases one unit apart in x: two cases that differ by one unit in x have mean outcomes β_1 apart, wherever they lie on the line. It compares different cases; it does not say what would happen to one case whose x changed. The intercept, $\beta_0 = 2$, is the mean of y at $x = 0$, the baseline for these comparisons. Reading it as a data-supported mean requires a meaningful zero within the predictor's support, as in this illustration but not for the occupations of Section 2.

In panel (c) each error ε_i is the vertical distance from a point to the true line, positive above it and negative below. Because the true line is unknown in practice, so are the errors; Section 2.3 measures distances from a fitted line instead, the residuals. Panel (d) shows what the error conditions claim: at every x the outcomes follow one Normal distribution centred on the line, with the same SD σ. Under Normal errors about two-thirds of the outcomes lie within σ of the

line and about 95% within 2σ. Here $\sigma = 1$, and four of the six invented points lie less than one unit from the line.

The line alone is a simple R graph; the surrounding scatter expresses the separate assumption about errors.

```
curve(2 + .5*x, from = -1, to = 5)
```

1.3 A process with known values

A simulation makes the model concrete. We choose the three parameters ourselves, generate data from the model, fit a line and compare the estimates with the values we chose. Take 20 predictor values equally spaced from 6 to 16, close to the range of the occupations' education in Section 2, and generate

$$Y_i = 2 + 0.5x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} \mathcal{N}(0, 1).$$

The true intercept is 2, the true slope 0.5 and the error SD 1; these are a teaching construction, not records of any study.

```
set.seed(123)
known_data <- data.frame(x = seq(6, 16, length.out = 20))
known_data$y <- 2 + 0.5 * known_data$x + rnorm(20, sd = 1)
known_model <- lm(y ~ x, data = known_data)
```

The predictor values and error SD are fixed by the generating process; the realized errors determine this sample's fitted line. Least squares, the rule of Section 2.4, supplies its coefficients. Table 1 compares the estimates with the known parameters.

Table 1
One fitted sample from a process with known coefficients and error SD.

Quantity	Estimate	Model SE
Intercept	2.4770	0.8362
Slope	0.4695	0.0733
Residual SD	0.9945	

The SE column reports uncertainty for the two coefficient estimates; the residual SD also has estimation uncertainty, which this table does not report. Section 3.1 derives the coefficient SEs, carrying the standard errors of Chapters 3 and 5 over to a line.

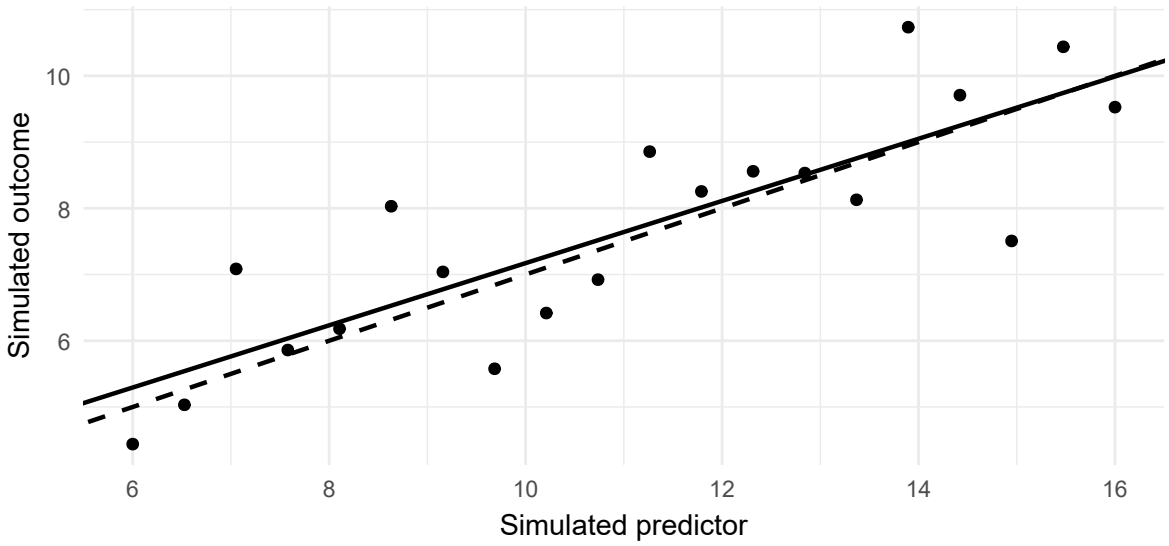


Figure 2: Known and fitted lines for one simulated dataset. The true process is programmed; the fitted line responds to the realized errors.

In this one sample each coefficient misses its programmed value by less than one SE: the slope is 0.4695 against 0.5, with SE 0.0733, and the intercept 2.477 against 2, with SE 0.8362. Misses of that size are usual. For this 20-point design about two-thirds of estimates fall within one SE of the truth and about 94% within two, the chances .6694 and .9392 that a t variable on 18 df lies within 1 and 2 of zero; Section 4 checks this by repeating the sample many times. In Figure 2 the errors are the vertical distances to the dashed true line, and the residuals of Section 2.3 the distances to the solid fitted line.

2 The Occupation Records and the Fitted Line

We now return to the occupations of the opening question.

2.1 The occupation records

The data are distributed as *carData::Prestige* (Fox et al., 2022; Fox & Weisberg, 2019). The package documentation attributes education and income to the *Census of Canada* (1971), volume 3, part 6, pages 19-1 to 19-21. We treat that documentation as a source record, not as evidence that the occupations form a probability sample, and the calculations are our reanalysis.

Table 2
Record, measurements, and scope for the focal analysis.

Feature	Meaning in this chapter
Unit and record count	102 occupations; each occupation receives equal weight
Education, x	Average years of education of occupational incumbents in 1971
Income, y	Average income of incumbents, divided by 1,000; thousand 1971 Canadian dollars
Missingness for the fit	No missing education or income values; no modeled rows excluded
Observed support	Education: 6.38–15.97 years; income: 0.611–25.879 thousand dollars
Descriptive estimand	Least-squares slope summarizing these 102 fixed occupation records
Limits of the source record	Cross-sectional and observational; no probability-sampling basis or individual-level analysis established here

Each occupation counts once, however many people it employs, so the analysis does not describe an average worker's experience; and 1971 dollars are not today's purchasing power.

2.2 Plot the records, then fit the line

```
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
library(ggplot2)
```

Dividing income by 1,000 expresses each occupation's recorded average in thousands of dollars; the occupations and their relative incomes stay the same.

```
ggplot(prestige, aes(education, income_k)) +
  geom_point(shape = 1, size = 2) +
  labs(x = "Average education (years)",
       y = "Average income (thousand 1971 dollars)")
```

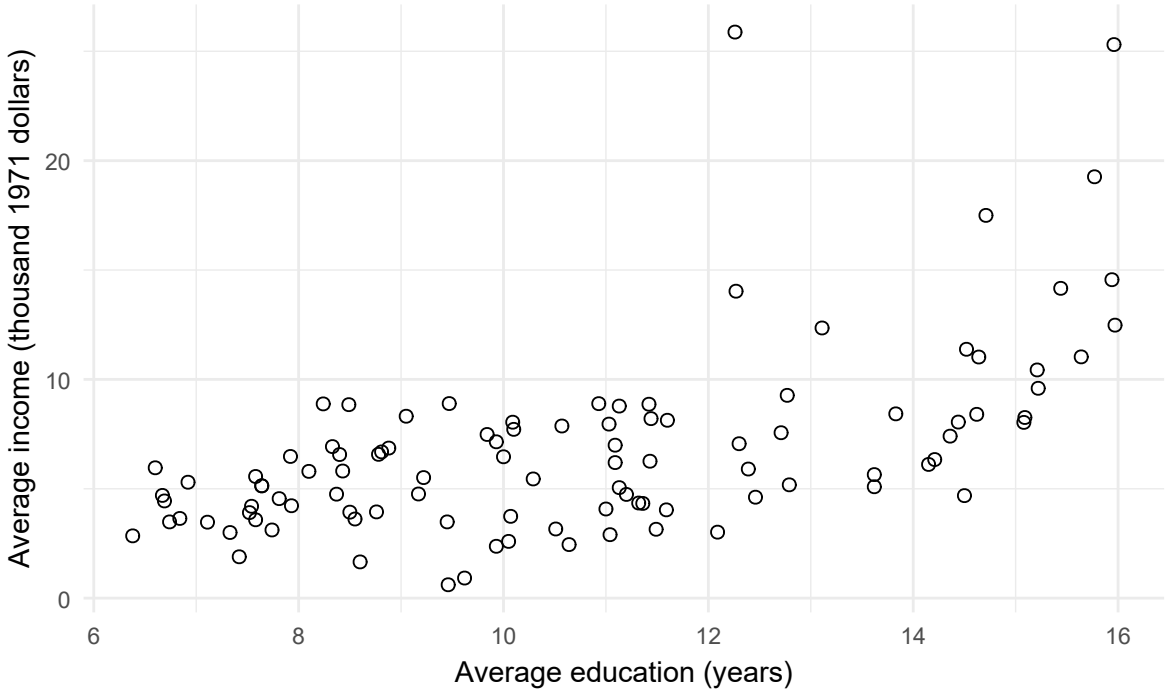


Figure 3: Education and income in the 102 occupation records, before fitting a line.

The scatterplot is panel (a) of Figure 1 without its line: the records are visible, but the process that produced them, if any, is not. Income rises with education, with much scatter. General managers average 12.26 years of education and 25.879 thousand dollars; university teachers average 15.97 years and 12.480. More education does not order every pair of occupations by income, so a line can summarize the overall pattern but cannot describe every occupation equally well. We save the straight-line fit as `model` and its coefficients as `b`.

```
model <- lm(income_k ~ education, data = prestige)
b <- coef(model)
coef(model)
```

```
#>   (Intercept)     education
#> -2.8535855824  0.8988128418
```

`lm()` finds the coefficients; it does not choose a population, establish a causal design or justify a confidence interval.

2.3 Fitted values, slope and residuals

A straight line assigns each observed predictor value x_i a **fitted value** $\hat{y}_i$, the line's prediction for record i:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad e_i = y_i - \hat{y}_i.$$

The subscript i indexes a record and the hat marks a fitted quantity. The **residual** e_i is observed minus fitted income: positive for an occupation above the line, negative below it. It is computed from the fitted line, unlike the error ε_i of Section 1, measured from the unknown true line.

For the occupation records, the line is

$$\widehat{\text{income}} = -2.8536 + 0.8988\,\text{education}.$$

With income in thousands of dollars, the slope says that fitted income differs by 0.8988 thousand dollars, about 899 dollars, per one-year difference in occupation-average education; a two-year difference corresponds to $2\hat{\beta}_1 \approx 1.7976$ thousand dollars.

For university teachers, the fitted coefficients before rounding give

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(15.97) \approx 11.5005, \qquad e = 12.4800 - \hat{y} \approx 0.9795,$$

so their recorded income lies about 980 dollars above their prediction. `predict()` applies the saved line to a record or to a new education value without refitting:

```
teacher <- prestige["university.teachers", ]
teacher_fit <- predict(model, teacher)
c(fitted = unname(teacher_fit),
  residual = teacher$income_k - unname(teacher_fit),
  at_12 = unname(predict(model, data.frame(education = 12))))
#>        fitted      residual         at_12
#> 11.5004555017  0.9795444983  7.9321685196
```

An occupation averaging 12 years of education has fitted income 7.9322 thousand dollars. Table 3 sets the slope's reading beside two sentences a report might also write about it.

Table 3
Three readings of one slope; only the first is a comparison these records make.

A sentence about the slope 0.8988	What it compares	Do these records support it?
Occupations whose average education is one year higher have fitted average income about 899 dollars higher	Different occupations, along the fitted line	Yes, as a description of the 102 records over 6.38–15.97 years
If an occupation's average education rose by one year, its average income would rise by about 899 dollars	One occupation before and after a change	No: it needs records of occupations over time and a design that isolates the change
Another year of schooling raises a worker's income by about 899 dollars	One person before and after more schooling	No: it needs records of individuals and a causal design; occupation averages cannot supply either

Only the first sentence compares what the records compare: occupations that differ in average education. The second describes a change within one occupation over time and the third a change within one person; the records observe neither (Robinson, 1950). The wording of a question signals which reading it asks for. Words such as *difference*, *relationship*, *higher*, *lower* and *predict* usually ask for a comparison between units; *effect*, *impact*, *change*, *increase* and *decrease* usually ask what a change would do, which needs a design that supports it. The opening report, "another year of education increases income", uses the second kind of wording for a comparison of the first kind.

`lm()` returns this line whether or not Section 1's model holds for the occupations. Here the line describes the 102 records: its slope, fitted values and residuals are completed calculations on them. A standard error and an interval answer a further question, which Section 3.1 asks once the model is declared for the occupations.

The intercept, −2.8536, is the line's fitted income at zero years of education. Zero lies far outside the observed range, so, as Section 1.2 warned, the intercept only places the line; no recorded occupation has negative income. Section 2.5 moves the intercept to a supported education value.

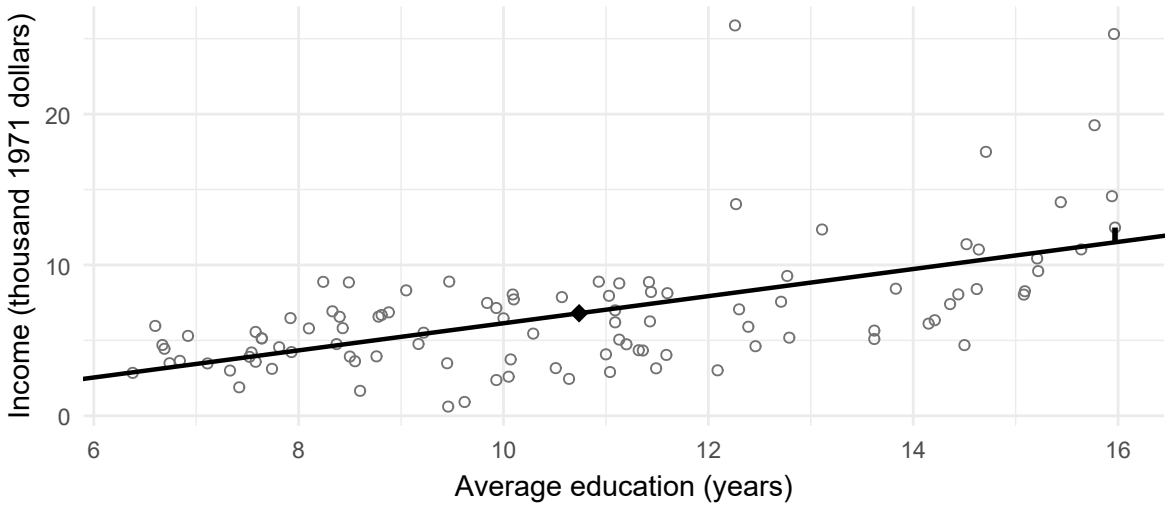


Figure 4: The least-squares line passes through the mean point; the marked vertical segment is the university-teacher residual.

2.4 How least squares chooses the line

Ordinary least squares (**OLS**) chooses the intercept and slope that minimize the **residual sum of squares**:

$$RSS = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

Squaring stops positive and negative residuals from cancelling and gives large residuals more weight. "Best" means best by this criterion among straight lines, not best for every research purpose. Two centered sums give the solution:

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \qquad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}),$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Here $\bar{x}, \bar{y}$ are means, s_x, s_y sample standard deviations, s_{xy} the sample covariance and r Pearson's correlation. The variance and covariance both divide by $n-1$, which cancels in their ratio. The slope carries thousand dollars per education year; the correlation has no units. For these data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} \approx 0.8988, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx -2.8536.$$

The scalar covariance is $s_{xy} = 6.6911$ thousand-dollar-years and the education variance is $s_x^2 = 7.4444$ years squared. Dividing them leaves thousand dollars per education year. Alternatively, the unit-free correlation $r = .5776$ multiplied by the income SD over the education SD restores the same slope and units. R applies both identities directly to the recorded education and income values:

```
x <- prestige$education
y <- prestige$income_k
cov(x, y) / var(x)
cor(x, y) * sd(y) / sd(x)
```

Both routes give 0.8988 using unrounded quantities from the same 102 complete pairs. This common set of records matters: separately calculated summaries that omit different occupations need not reproduce the fitted line.

A slope needs education to vary: if every occupation had the same education, $S_{xx} = 0$ and no slope could be identified. The line passes through $(\bar{x}, \bar{y})$, since substituting $\bar{x}$ into the intercept formula gives $\bar{y}$. With an intercept, the residuals also sum to zero and their products with x sum to zero; otherwise the line could be shifted or tilted to reduce RSS. These are algebraic properties of OLS, not tests of independence or linearity. Because they tie the residuals together, residuals are an imperfect window on the errors, which Section 6 keeps in mind. **Appendix A** derives them.

Three teaching records, $(x, y) = (1, 2), (2, 4), (3, 3)$, show the whole calculation at once. Their means are $\bar{x} = 2$ and $\bar{y} = 3$. Table 4 lays out the centered values, their squares and products, the fitted values and the residuals, with column sums in the last row.

Table 4
The complete least-squares calculation for three records; the column sums give S_{xx}, S_{xy} and RSS.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x-\bar{x})(y-\bar{y})$	$\hat{y}$	e	e^2
1	2	−1	−1	1	1	2.5	−0.5	0.25
2	4	0	1	0	0	3.0	1.0	1.00
3	3	1	0	1	0	3.5	−0.5	0.25
Sum		0	0	2	1	9.0	0.0	1.50

The sums give $S_{xx} = 2$ and $S_{xy} = 1$, so

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{1}{2} = 0.5, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3 - 0.5(2) = 2.$$

The fitted values 2.5, 3.0 and 3.5 lie on this line, the residuals sum to zero, and their squares sum to $RSS = 1.5$. No Normality assumption is needed; the model of Section 1 enters only when we ask about repeated-sample uncertainty.

2.5 Candidate lines and centering

Least squares also lets us compare the fitted line with alternatives. Make every candidate pass through $(\bar{x}, \bar{y})$ and vary only its slope b_1; then

$$RSS(b_1) = RSS(\hat{\beta}_1) + S_{xx}(b_1 - \hat{\beta}_1)^2.$$

The added term is zero only at the OLS slope. Table 5 and Figure 5 check five lines on the occupation records; they are alternative summaries of the same records, not repeated samples.

Table 5
Candidate lines through the mean point and their residual sums of squares.

Intercept	Slope	RSS
3.5892	0.2988	1484.0707
0.3678	0.5988	1281.0617
−2.8536	0.8988	1213.3920
−6.0750	1.1988	1281.0617
−9.2964	1.4988	1484.0707

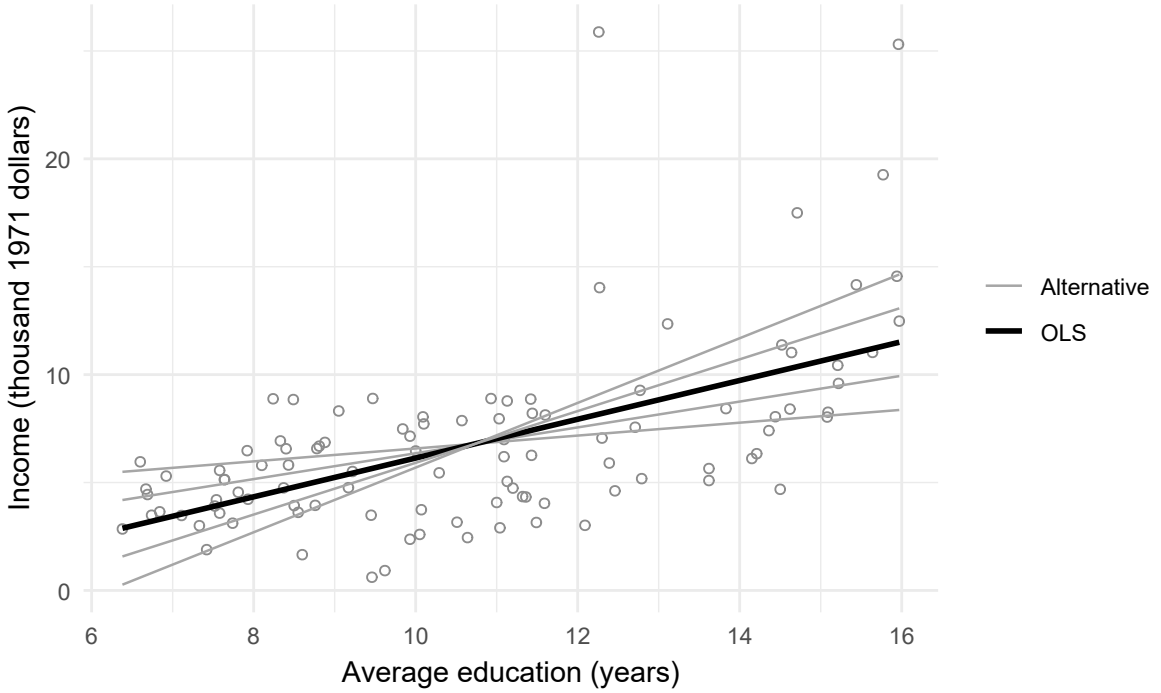


Figure 5: Five candidate lines through the sample mean. The OLS line has the smallest residual sum of squares.

Every alternative has a larger RSS than the OLS line, and slopes equally far above and below 0.8988 add the same excess. Because large vertical discrepancies weigh heavily, high-income occupations can pull the line, a feature of the criterion rather than evidence that they are wrong. The squared residuals of the fitted model recover its minimum RSS:

```
sum(resid(model)^2)
```

The fitted line can also be written around a chosen reference point:

$$\hat{y}_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x}).$$

Centering education at its mean makes the intercept $\bar{y} = 6.7979$ thousand dollars, the fitted income at average education; centering at 12 makes it 7.9322. The slope, fitted values, residuals and R^2 stay the same: centering moves the reference point, not the line. Changing income from thousands of dollars to dollars multiplies the intercept, slope, residual SD and the coefficient SEs of Section 3.1 by 1,000 and leaves the t and p of Section 3.1, r and R^2 unchanged; a unit change is not new evidence.

2.6 How well does the line fit?

A line can show a clear trend and still leave large differences between occupations. For OLS with an intercept, the total variation of income around its mean splits into two sums of squares:

$$TSS = \sum_i (y_i - \bar{y})^2 = RegSS + RSS, \qquad RegSS = \sum_i (\hat{y}_i - \bar{y})^2.$$

For the occupation records, $1820.8134 = 607.4214 + 1213.3920$, in squared thousand dollars. These are sums of squared deviations, not variances; the split holds because the fitted deviations and the residuals have a zero cross-product. The coefficient of determination is

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{1213.3920}{1820.8134} \approx .3336.$$

The line accounts for a proportion .3336 of the squared variation of income around its mean. "Accounts for" is algebraic: it does not mean that education causes or determines income. With an intercept and one predictor, $R^2 = r^2$, here $.5776^2 \approx .3336$.

The residual standard deviation, in thousand dollars, is

$$s_e = \sqrt{\frac{RSS}{n-2}} = \sqrt{\frac{1213.392}{100}} = 3.4834.$$

Fitting an intercept and a slope leaves $n - 2$ residual degrees of freedom. Under the model of Section 1.1, s_e^2 is an unbiased estimate of the error variance and s_e estimates the error SD σ (Appendix A); without that model, s_e measures residual scatter. It is not the slope's standard error. `sigma()` returns it, and `summary()` reports R^2:

```
c(residual_SD = sigma(model), R_squared = summary(model)$r.squared)
#>  residual_SD    R_squared
#> 3.4833777071 0.3335989192
```

Table 6 sets four summaries side by side. Here and below, SE denotes an estimated standard error; the last row needs Section 1's model declared for the occupations, which Section 3.1 does, and is not part of the completed calculation on the 102 records.

Table 6
Different summaries have different units and jobs.

Quantity	Occupation result	What it describes
Slope	0.8988	Change in fitted thousand-dollar income per education year
Residual SD	3.4834	Remaining income scatter on the outcome scale
R^2	.3336	Proportion of centered squared variation accounted for in these records
Classical slope SE	0.1270	Model-based repetition variability of the slope, once Section 1's model is declared (Section 3.1)

3 Inference for the Slope

3.1 The slope's standard error and confidence interval

The slope 0.8988 is not waiting to be discovered: we computed it from all 102 records named in the question. A confidence interval answers a further question, about a process that could generate new outcomes. Apply the model of Section 1.1 to the occupations: suppose each occupation's income arose independently from a process whose mean at education x_i is $\beta_0+\beta_1 x_i$, with errors of common variance σ^2, Normal for the exact t procedure. The estimand is now β_1, the slope of that hypothetical process, not the fixed slope of the records, and repetition means new incomes at the same 102 education values, not a new sample of occupations. Nothing in the records establishes this process: it is declared, and its conditions are a proposed justification that Section 6 examines.

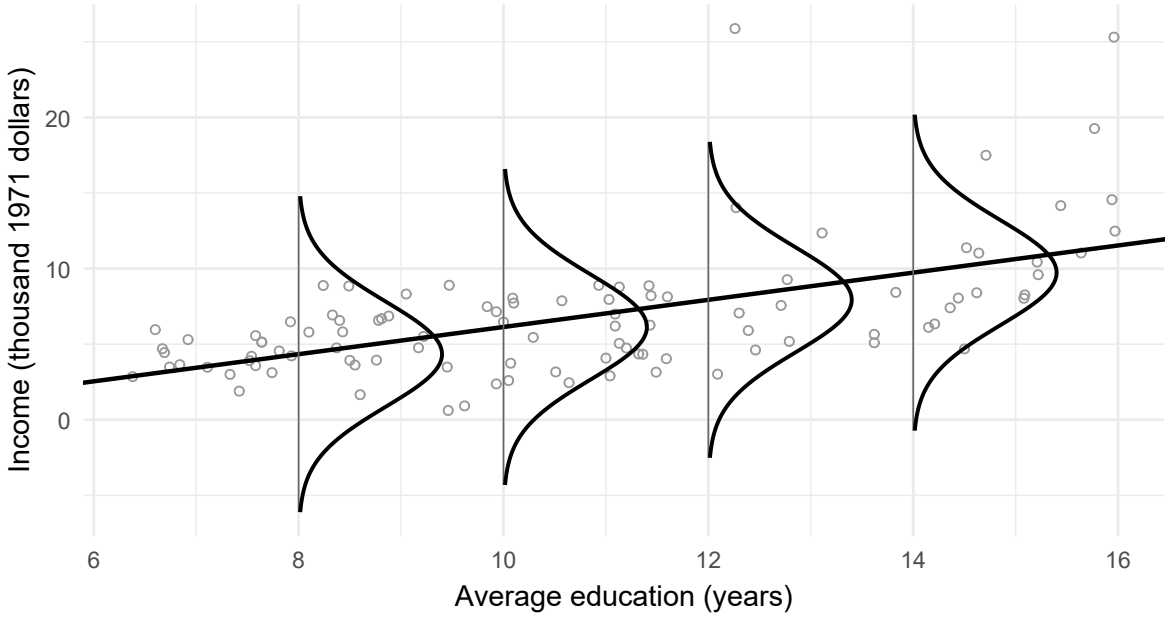


Figure 6: The model of Section 1 drawn on the occupation records, with the fitted line and the residual SD in place of the unknown parameters. The records depart from it: their spread grows with education, and the curve at 8 years reaches below zero income.

Figure 6 draws that model on the records, as panel (d) of Figure 1 drew it on invented points, with the fitted line and $s_e = 3.4834$ in place of the unknown β_0, β_1 and σ. The records do not look like that everywhere. At higher education they spread more widely, and several incomes lie beyond the upper tails; at 8 years the curve reaches below zero, a sign that a Normal model for a nonnegative outcome is an approximation, which Section 5 meets again.

Substituting the process into the slope formula shows where slope uncertainty comes from. The observed predictor values stay fixed, as in Section 1.1, and $\sum_i (x_i - \bar{x}) = 0$, so

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x})\varepsilon_i}{S_{xx}}.$$

The fitted slope misses β_1 because the realized errors need not cancel once weighted by distance from mean education. Independence and common variance give

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2 \sum_i (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}},$$

which needs no Normality; Normality enters the exact t reference.

Estimating σ by s_e gives the standard error:

$$SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{S_{xx}}} = \frac{3.4834}{\sqrt{751.8852}} = 0.127.$$

The 95% confidence interval is

$$\hat{\beta}_1 \ \pm \ t_{.975,n-2} SE(\hat{\beta}_1) \approx 0.8988 \pm 1.984(0.127) \approx [0.6468,\ 1.1508].$$

Here 1.984 is the 97.5th percentile of t with 100 degrees of freedom. More error variation raises the SE, and a wider predictor spread lowers it.

The education values `x` defined in Section 2.4 give the same SE:

```
sigma(model) / sqrt(sum((x - mean(x))^2))
```

The test of $H_0 : \beta_1 = 0$ gives $t = \hat{\beta}_1 / SE(\hat{\beta}_1) \approx 7.0753$ and a two-sided $p < .0001$: evidence against the zero-slope model under the stated procedure. It is not the probability that the null is true, evidence of causation or a measure of practical importance. The interval describes what the procedure achieves over repetitions, which Section 4 makes visible, not a 95% probability for the fixed parameter. The coefficient table gives the estimate, SE and test, and `confint()` the interval, at its default level .95:

```
coef(summary(model))
#>                  Estimate   Std. Error      t value        Pr(>|t|)
#> (Intercept) -2.8535855824 1.4070392470 -2.028078171 4.521163138e-02
#> education    0.8988128418 0.1270354112  7.075293679 2.079192410e-10
confint(model, "education")
#>                  2.5 %      97.5 %
#> education 0.6467782042 1.150847479
```

3.2 The slope test and the correlation test

Chapter 5 tested a zero Pearson correlation. For an OLS line with an intercept, fitted to the same complete pairs,

$$t_{\text{slope}} = \frac{\hat{\beta}_1}{s_e / \sqrt{S_{xx}}} = r\sqrt{\frac{n-2}{1-r^2}} = t_{\text{Pearson}},$$

$$F_{\text{regression}} = t^2_{\text{slope}} = \frac{R^2}{1-R^2}(n-2).$$

The first identity follows from the covariance form of the slope and $RSS = (1 - r^2)TSS$; the second compares the intercept-only model with the line, and squaring the signed t gives the nondirectional F. For the occupation line both t statistics are 7.0753 on 100 df, and the F statistic that `summary()` prints is $F(1, 100) = 50.0598 = t^2$, all with $p < .0001$. They are one comparison written three ways, not three pieces of evidence.

The equality is algebra, and it needs the same intercept, rows, weights and coefficient-covariance convention: a robust slope test of the kind Section 6 introduces, which estimates the slope's SE without assuming equal error variance, need not equal the ordinary F. For randomly sampled bivariate-Normal pairs a zero population correlation, $\rho = 0$, also means a zero conditional-mean slope, but the occupation records have no sampling warrant, so here the Pearson test only reproduces the process calculation.

3.3 A binary predictor and the two-sample t test

A predictor need not be continuous. Chapter 4's invented feedback study gave each of its 24 students a score out of 20, twelve under massed and twelve under spaced practice. Code the schedule $x_i = 0$ for massed and $x_i = 1$ for spaced, and fit the same kind of line. With only two predictor values, the least-squares line joins the two group means (Figure 7).

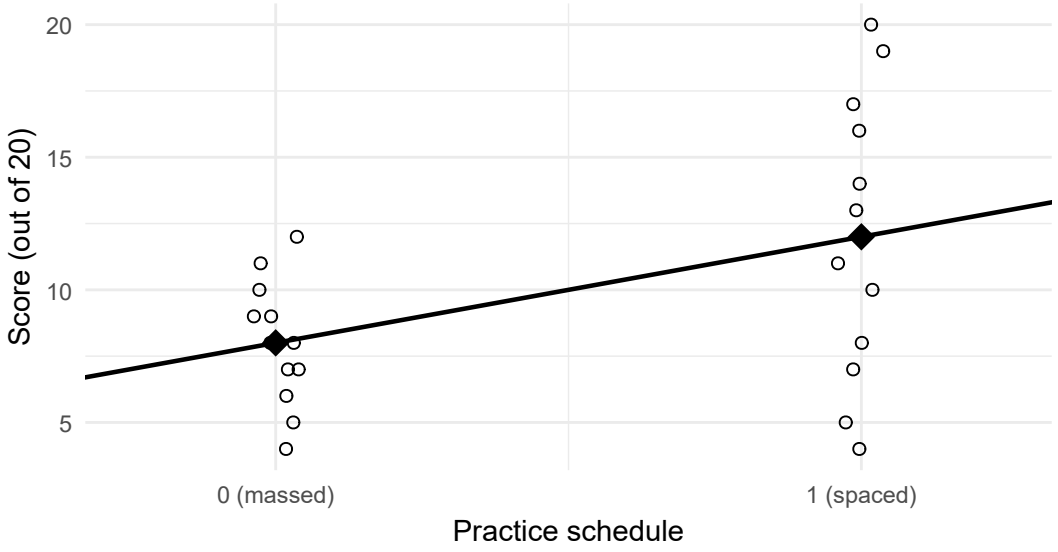


Figure 7: With a 0/1 predictor, the least-squares line joins the two group means: the intercept is the massed mean and the slope the difference in means.

Its coefficients take the readings of panel (b) of Figure 1: the intercept is the mean at $x = 0$, the massed mean, and the slope the difference for a one-unit step in x, from massed to spaced,

$$\hat{\beta}_0 = \bar{y}_0 = 8, \qquad \hat{\beta}_1 = \bar{y}_1 - \bar{y}_0 = 12 - 8 = 4.$$

The residual SD is the pooled SD, $s_e = 4.156$ on $24 - 2 = 22$ df, and the slope SE becomes $SE(\hat{\beta}_1) = s_e\sqrt{1/n_0 + 1/n_1} = 4.156\sqrt{1/12 + 1/12} = 1.6967$, so $t = 2.3575$ on 22 df, $p = .0277$ and the interval is $[0.4813,\ 7.5187]$. These are exactly the equal-variance two-sample t test and interval, written as a line, with $R^2 = .2017$ and $F(1, 22) = 5.5579 = t^2$.

```
score <- c(4, 5, 7, 8, 4, 5, 7, 8, 6, 7, 9, 10, 10, 11, 13, 14,
           8, 9, 11, 12, 16, 17, 19, 20)
spaced <- rep(rep(c(0, 1), each = 4), 3)
two_groups <- lm(score ~ spaced)
coef(summary(two_groups))
```

Welch's test uses the separate group SDs, 2.3741 and 5.3767, and gives 15.1322 df and $p = .0323$; the classical slope SE and test above instead use one common error variance. The HC3 covariance of Section 6 keeps the line and gives a slope SE of 1.7721. A factor with more than two levels needs more than one 0/1 column; Chapter 9 develops that coding, its contrasts and the link to the analysis of variance of Chapter 4.

4 Slopes Across Repeated Samples

4.1 Repeated samples and coverage

Section 1.3 drew one sample of 20 from the known process $Y_i = 2 + 0.5x_i + \varepsilon_i$ with $\sigma = 1$. Now fix 40 predictor values over the same range, 6 to 16, and draw 2,000 samples from the same generating rule. The study size is 40; the 2,000 repetitions only refine the simulation's description of that study. Each repetition draws new errors, adds them to the same 40 process means, fits a line, records its slope and SE, and checks whether its 95% interval contains 0.5.

Table 7

Slope behavior across 2,000 repetitions of the known process.

Quantity	Value
Mean estimated slope	0.5018
SD of estimated slopes	0.0528
Mean estimated SE	0.0531
Theoretical slope SD	0.0534
Fraction of intervals covering 0.5	.9455

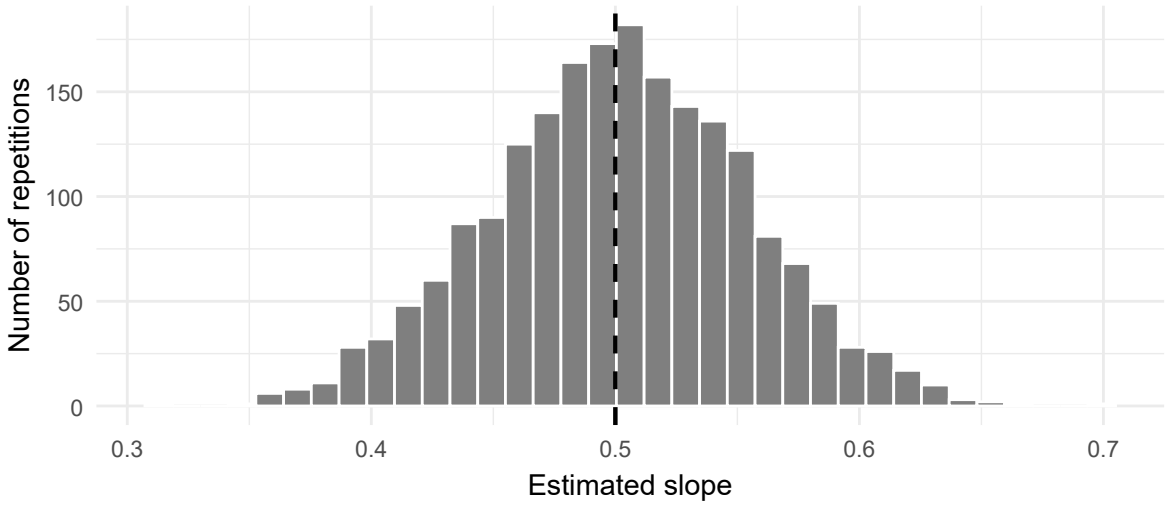


Figure 8: Slopes in 2,000 repetitions of the known process. The vertical reference is the true slope, 0.5.

The SD of the 2,000 slopes estimates their repeated-sample spread, and each fitted SE estimates that spread from one sample; on average the two agree closely, and both are close to the theoretical SD $\sigma/\sqrt{S_{xx}}$ that Section 3.1's variance formula gives with $\sigma = 1$. Measured in their own SEs, .6890 of the slopes lie within one SE of 0.5 and .9425 within two, close to the .6764 and .9473 of a t variable on 38 df: the reading of one sample in Section 1.3 holds across many. Coverage belongs to the interval procedure, not to one realized interval, and with 2,000 repetitions its Monte Carlo SE at a true .95 is $\sqrt{.95(.05)/2000} \approx 0.0049$, so the observed .9455 lies within one Monte Carlo SE of .95.

4.2 What changes slope precision

Table 8 uses the theoretical slope SD $\sigma/\sqrt{S_{xx}}$ to change one feature of the design at a time.

Table 8
How sample size, predictor support, and error SD affect theoretical slope precision.

Condition	n	Lower	Upper	Error SD	Slope SD
Baseline	40	6.0	16.0	1	0.0534
More rows	160	6.0	16.0	1	0.0272
Narrower support	40	8.5	13.5	1	0.1068
More noise	40	6.0	16.0	2	0.1068

Halving the predictor range or doubling σ doubles the slope SD; quadrupling the points over the same range about halves it, not exactly, because the two grids' variances differ slightly. Taking `x <- 1:n` with a growing n would change size and range together. More rows do not remove confounding, measurement error, dependence or misspecification; these simulations show behavior only under the mechanism programmed into them.

5 Prediction

At a chosen fixed education value x_0 the line gives one number, which can answer three questions of increasing uncertainty: what the line predicts, what the mean income of occupations at x_0 is, and what one new occupation's income will be. The last two need Section 1's model, declared for the occupations in Section 3.1.

5.1 Point prediction

The **point prediction** at x_0 is the fitted line's value,

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0,$$

built from the estimates, not the unknown parameters. At $x_0 = 12$ it is 7.9322 thousand dollars, the value `predict()` returned in Section 2.3. By itself a point prediction says nothing about its uncertainty, and the same number serves two different targets: it estimates the mean income of occupations at x_0 and predicts one new occupation's income.

5.2 The mean response and its confidence interval

Under the process model, occupations with education x_0 have mean income $\mu_0 = \beta_0 + \beta_1 x_0$, the **mean response**, a process parameter. The point prediction estimates it, $\hat{\mu}_0 = \hat{y}_0$, with standard error

$$SE(\hat{\mu}_0) = s_e\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = 3.4834\sqrt{\frac{1}{102} + \frac{(12 - 10.738)^2}{751.8852}} = 0.3803$$

at 12 years. The 95% **confidence interval for the mean response** is $7.9322 \pm 1.984(0.3803) \approx [7.1776,\ 8.6868]$. The SE is smallest at $\bar{x}$, grows with distance from it, and shrinks as independent records accumulate: it describes uncertainty about an average, not the spread of individual occupations. `predict()` returns the fit and the two limits:

```
predict(model, data.frame(education = 12), interval = "confidence")
#>          fit        lwr         upr
#> 1 7.93216852 7.17757965 8.686757389
```

5.3 A new case and its prediction interval

One new occupation at x_0 has income $Y_0 = \mu_0 + \varepsilon_0$, a random outcome with its own error. Predicting it by $\hat{\mu}_0$ makes an error with two parts,

$$Y_0 - \hat{\mu}_0 = \varepsilon_0 - (\hat{\mu}_0 - \mu_0),$$

the new case's own error and the error in the estimated mean. When the new error is independent of the fitted data and comes from the same common-variance process, their variances add:

$$\mathrm{Var}(Y_0 - \hat{\mu}_0) = \sigma^2 + \mathrm{Var}(\hat{\mu}_0).$$

Substituting the mean-response variance and estimating σ by s_e gives

$$SE_{\text{new}} = s_e\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = 3.5041,$$

and the 95% **prediction interval** $7.9322 \pm 1.984(3.5041) \approx [0.9802,\ 14.8842]$. The extra 1 is the new outcome's own error variance, in units of σ^2; unlike the mean-response SE, it does not shrink as the estimation sample grows.

```
predict(model, data.frame(education = 12), interval = "prediction")
#>          fit          lwr        upr
#> 1 7.93216852 0.9801724429 14.8841646
```

5.4 Confidence and prediction intervals compared

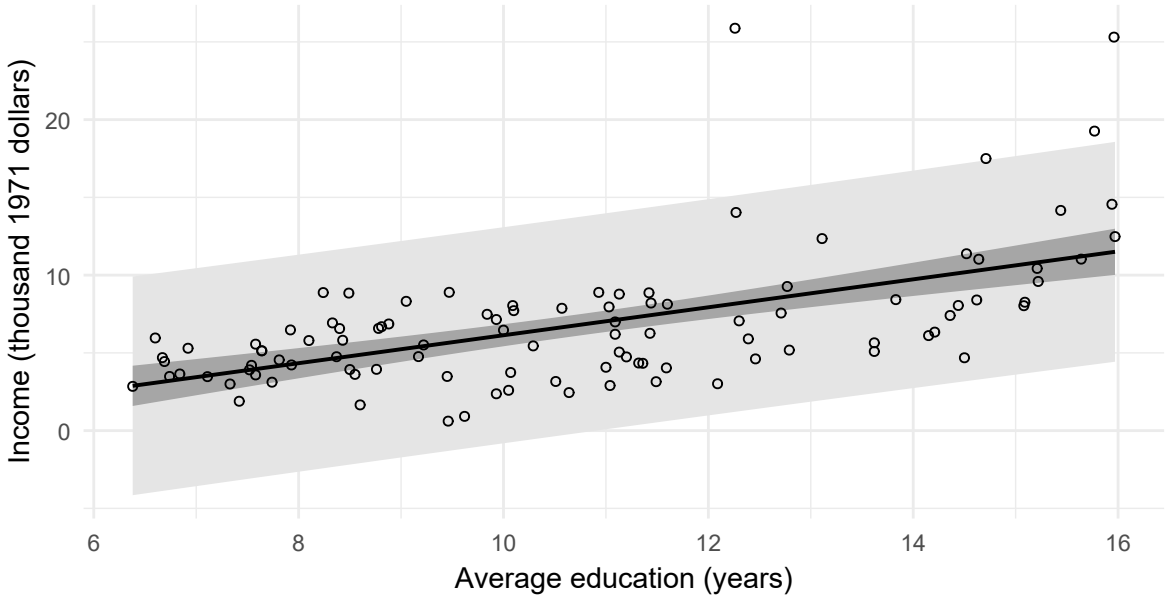


Figure 9: Confidence intervals for the mean response (dark band) and prediction intervals for one new occupation (light band), from the same fitted line.

Figure 9 draws both intervals across the observed education range. The prediction band is much wider everywhere, because it carries a new occupation's own scatter as well as the uncertainty about the line; the confidence band carries only the latter. Both widen away from $\bar{x}$. Below about 11 years the prediction band's lower limit falls below zero, which exposes the limits of a Normal model for a nonnegative outcome rather than forecasting negative income. Outside 6.38–15.97 years the software still returns numbers, but the records cannot check them. Neither interval measures how well the line predicts records it has not seen; that held-out question belongs to Chapter 8. Appendix A derives the mean-response variance from the coefficient variances and their covariance, and gives both intervals at four education values.

6 Diagnostic Checks

6.1 Residuals, influence and unequal spread

The fitted line exists whether or not the model of Section 1.1 holds. Claims that it describes a linear conditional mean, that its interval has the stated coverage, or that it predicts new occupations well need more support, and a check earns its place when its result could change one of those claims.

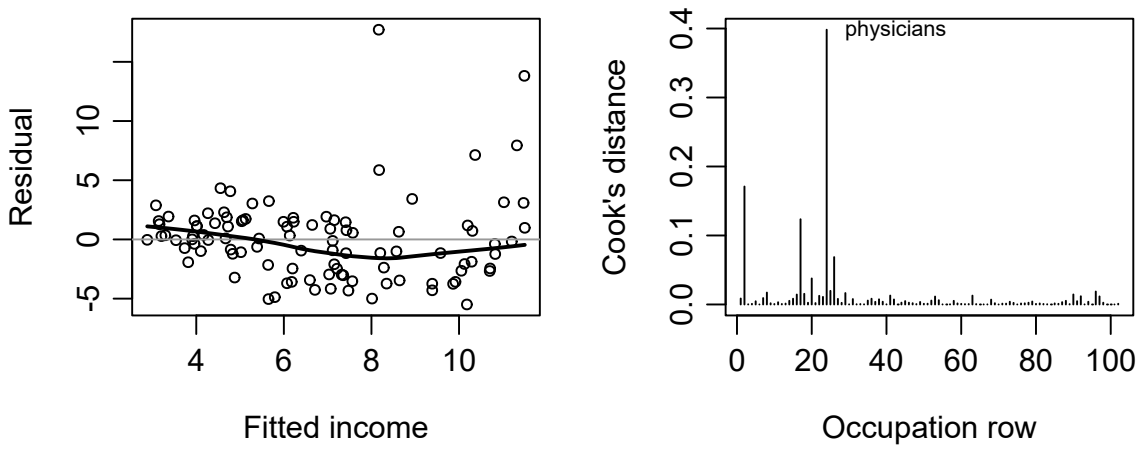


Figure 10: Residual pattern and influence invite different follow-up questions.

A curve in the residuals (Figure 10, left) suggests that one straight conditional mean leaves systematic variation, and a residual spread that changes with the fitted value challenges the common-variance SE. Leverage concerns an unusual predictor value; influence concerns how much the fit changes when a record is left out. A large residual, high leverage and high influence are related but different properties.

A minimal version of the first display plots residuals against fitted values, with a horizontal reference at zero:

```
plot(fitted(model), resid(model))
abline(h = 0, lty = 2)
```

For a straight line with an intercept, the leverage and Cook's distance of record i are

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}, \qquad D_i = \frac{e_i^2}{2s_e^2} \frac{h_i}{(1 - h_i)^2}.$$

Leverage grows with distance from $\bar{x}$; Cook's distance combines it with the residual to summarize how far the fitted values move when record i is omitted (Figure 10, right). It prompts a look at provenance and a sensitivity analysis, not automatic deletion.

Adding education squared lets the fit bend. This same-record comparison is an exploratory check, not a new final model:

```
quad <- lm(income_k ~ education + I(education^2), data = prestige)
most <- which.max(cooks.distance(model))
without <- lm(income_k ~ education, data = prestige[-most, ])
c(full = unname(coef(model)[2]), without = unname(coef(without)[2]))
#>         full      without
#> 0.8988128418 0.7982204353
coef(summary(quad))["I(education^2)", ]
#>       Estimate     Std. Error        t value       Pr(>|t|)
#> 0.134179313419 0.047636405684 2.816738825952 0.005856544055
```

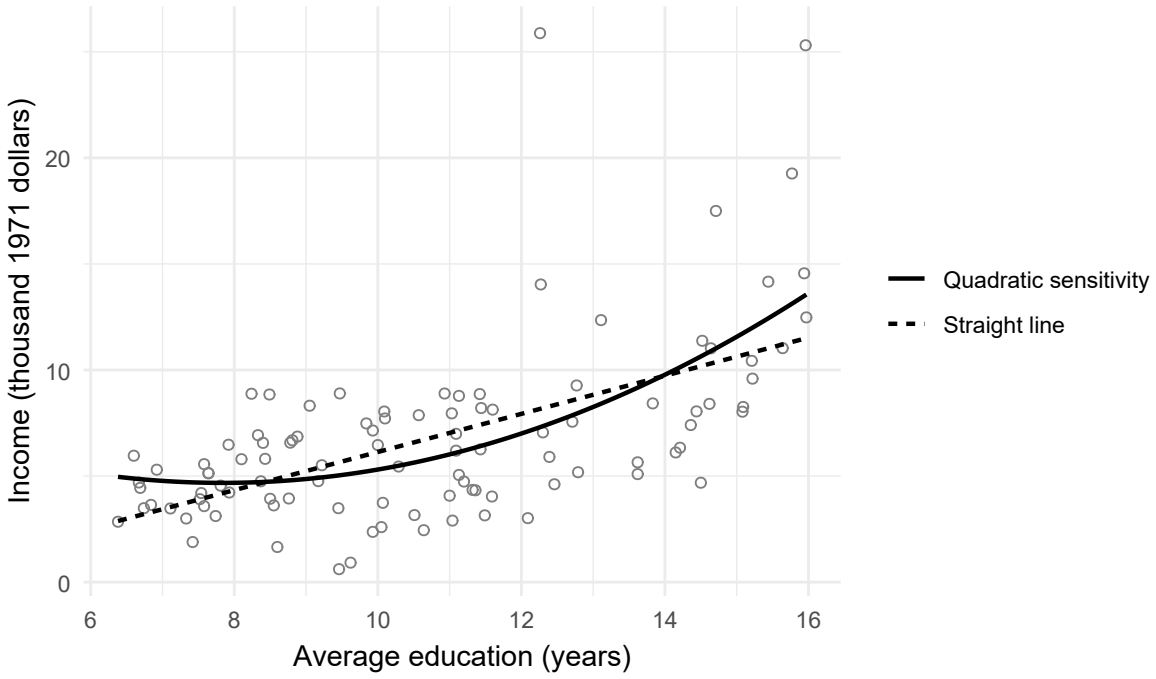


Figure 11: Straight and quadratic fits to the same 102 records. The curve is an exploratory sensitivity, not a selected final model.

The quadratic coefficient, 0.1342, has $p = .0059$ under that model's classical calculation: a reason to look at the curvature rather than a model-selection rule, since the spread is unequal and the comparison was chosen after seeing the records. Omitting physicians, the record with the largest Cook's distance (0.3983), moves the slope from 0.8988 to 0.7982. Nothing shows that physicians were recorded wrongly; the deletion is a sensitivity disclosure, not a replacement sample.

An HC3 heteroskedasticity-consistent covariance gives a slope SE of 0.1491 rather than 0.1270 and, with the same t_{100} multiplier, an approximate interval [0.6031, 1.1945]. The line does not change. HC3 relaxes the equal-variance assumption for independent records; it does not repair curvature, dependence, confounding or measurement error (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980; Zeileis, 2004).

```r
V_hc3 <- sandwich::vcovHC(model, type = "HC3")
c(classical = coef(summary(model))["education", "Std. Error"],
  HC3 = sqrt(V_hc3["education", "education"]))
#>    classical          HC3
#> 0.1270354112 0.1490510028
```

Later chapters go further: Chapter 8, Appendix B, builds the HC3 matrix from its parts; Chapter 9, Appendix A, defines what a least-squares line estimates when the conditional mean is not linear; and Chapter 14, Section 5.1, illustrates attenuation under independent classical predictor error in a single-predictor model. Nothing here relies on them. Table 9 collects the five checks and what each can change.

Table 9
Diagnostic questions and their consequences.

Check	What needs to be examined	Consequence
Form	Residual curve and a curved-fit comparison	A straight line may remain a description but fail as a constant conditional-mean slope account
Spread	Residual variation over fitted values	Classical SEs may need revision; HC3 is a variance sensitivity, not a universal repair
Influence	Source record and result with/without a consequential case	Disclose sensitivity; correct errors only when verified
Independence	How occupations and measurements were generated or related	A plot cannot establish independent process realizations
Support	Actual education range and density of observations	Restrict comparisons; avoid extrapolation and individual-level claims

7 Regression Toward the Mean

Regression toward the mean first describes a simple line's predictions. In standard units, with an imperfect correlation, its prediction for a second measurement lies nearer its mean than the first measurement lies to its own. Group averages follow that pattern when the conditional mean is approximately linear; individual outcomes and particular groups still vary around their predictions. The phenomenon comes in two versions. In the first, two different but related quantities are measured, such as a father's height and his son's (Section 7.1). In the second, one quantity is measured twice, with some luck in each measurement, such as a student's score on two exams (Section 7.3). Either way the movement runs in both directions, as Section 7.2 shows by selecting on the sons as well as on the fathers.

7.1 Version one: fathers' and sons' heights

Francis Galton, who gave regression its name, noticed in the 1880s that the grown children of tall parents were taller than average but, on average, less tall than their parents, and the children of short parents less short (Galton, 1886; Gelman et al., 2020). He measured the adult heights of parents and their grown children in 205 families. We use James Hanley's transcription of Galton's notebook (Hanley, 2004), distributed in the R package HistData (Friendly et al., 2025), and keep one row per family: among the 179 families with at least one son, one son was drawn at random from each, so every father appears once and the rows describe separate families rather than brothers who share a father. Heights are in inches, mostly recorded to the nearest half inch. The small extract `galton1886_sons.csv` is supplied with the chapter under this note: *Galton's family heights (published 1886), transcribed by J. A. Hanley and distributed in the R package HistData under the GPL; one son per family drawn for QMSBR. The measurements are historical records in the public domain; the extract is redistributed with this attribution.*

Fathers average 69.0989 inches and sons 69.2140. Written around the fathers' mean, in the centred form of Section 2.5, the least-squares line that predicts a son's height from his father's is

$$\widehat{\text{son}} = 69.2140 + 0.4438\,(\text{father} - 69.0989).$$

A father of average height is predicted to have a son of average height, and each inch a father stands above or below the fathers' mean predicts about 0.44 inch above or below the sons' mean. As in Section 2.3, the slope compares families: fathers an inch apart have sons predicted 0.4438 inch apart. (The uncentred intercept, 38.5445 inches, is the prediction for a father of zero height and places the line only.) Because the two generations' SDs differ slightly, the general statement uses standard units. Chapter 2 measured a value's distance from its mean in standard units, $z_x = (x - \bar{x})/s_x$, and Section 2.4 wrote the slope as $\hat{\beta}_1 = r\, s_y/s_x$. Substituting

that slope into the line through the mean point, $\hat{y} - \bar{y} = \hat{\beta}_1(x - \bar{x})$, and dividing by s_y gives the line in standard units:

$$\frac{\hat{y} - \bar{y}}{s_y} = r\,\frac{x - \bar{x}}{s_x}, \qquad \text{that is,} \qquad \hat{z}_y = r\,z_x.$$

The line passes through the origin with slope r. Unless the correlation is perfect, $|r| < 1$, so a case z_x SDs from the mean on the predictor is predicted only $r\,z_x$ SDs from the mean on the outcome. Galton called this regression toward mediocrity; we call it regression toward the mean.

```
galton <- read.csv("galton1886_sons.csv")
sapply(galton[c("father", "son")], sd)
#>      father         son
#> 2.546555039 2.647465487
c(r = cor(galton$father, galton$son),
  z_slope = unname(coef(lm(scale(son) ~ scale(father),
    data = galton))[2]))
#>           r      z_slope
#> 0.4269305719 0.4269305719
```

The fathers' SD is 2.5466 inches and the sons' 2.6475, and $r = .4269$. In inches the fitted slope is 0.4438 inch of son's height per inch of father's height; in standard units it is .4269, which is r. A father 1.6 SD above the fathers' mean has a son predicted $r(1.6) \approx 0.6831$ SD above the sons' mean: taller than average, but less than half as far out.

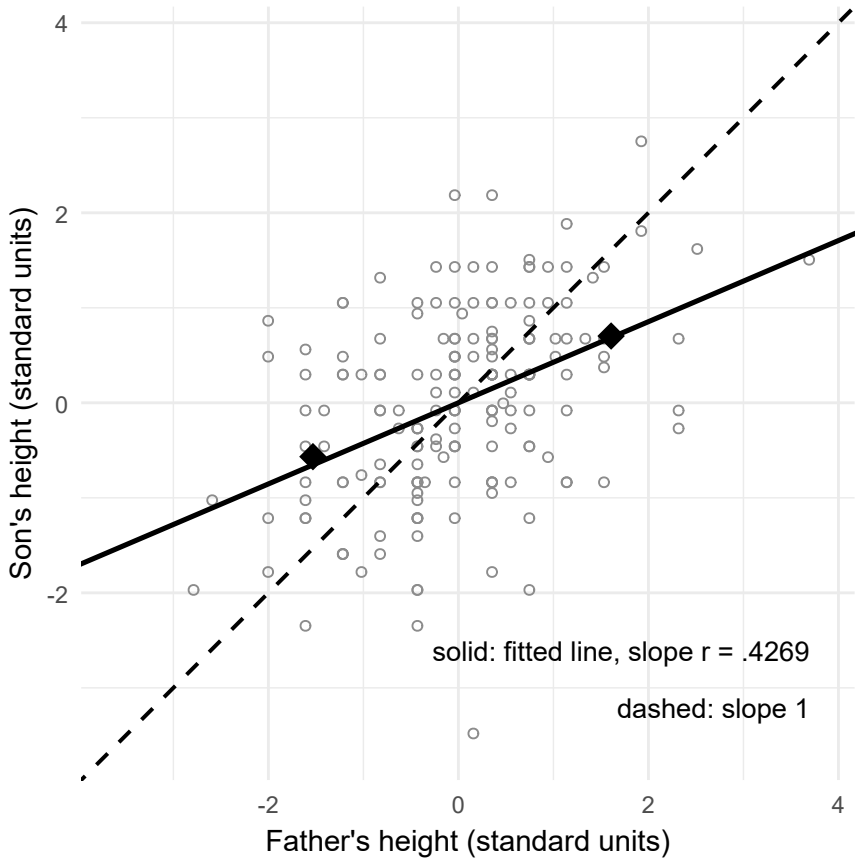


Figure 12: Galton's 179 fathers and sons in standard units. The fitted line has slope $r = .4269$, flatter than the dashed line of slope 1; the filled points are the mean positions of the fathers more than 1 SD above and below their mean, with their sons.

In Figure 12 the dashed line of slope 1 marks where a son would sit if he were exactly as many SDs from his mean as his father. The fitted line is flatter, lying below the dashed line to the right of the mean and above it to the left: tall fathers have sons predicted less tall, and short fathers sons predicted less short, in standard units. The two filled points, the average positions of the fathers more than 1 SD above and below their mean with their sons, sit near the fitted line, well inside the dashed one.

7.2 Extreme groups and the regression paradox

Table 10 groups the men in bands of 1 SD, first by the fathers' heights and then by the sons'. Each band includes its upper bound, so the four bands of each block hold all 179 men.

Table 10
Groups of Galton's fathers and sons in standard units, selected on either height.

Selected on	Band (SDs)	Men	Mean z of the selected height	Mean z of the other height	r times the selected mean
Fathers	Above +1	24	1.6072	0.7029	0.6862
Fathers	0 to +1	60	0.4978	0.2251	0.2125
Fathers	−1 to 0	65	−0.3463	−0.2058	−0.1479
Fathers	Below −1	30	−1.5310	−0.5668	−0.6536
Sons	Above +1	31	1.4057	0.6224	0.6001
Sons	0 to +1	60	0.4650	0.1431	0.1985
Sons	−1 to 0	61	−0.4610	−0.0697	−0.1968
Sons	Below −1	27	−1.6057	−0.8751	−0.6855

The 24 fathers more than 1 SD above their mean average $z = 1.6072$, or 73.1917 inches; their sons average 0.7029, or 71.0750 inches, close to the line's prediction $r(1.6072) \approx 0.6862$. The 30 short fathers mirror them, and in every row the other height's mean lies between zero and the selected mean, near r times it. Selecting on the sons gives the same movement backward in time: the 31 sons more than 1 SD above their mean, at 1.4057, had fathers averaging only 0.6224. No force pulls heights toward the average; a force could not act backward in time. The imperfect correlation guarantees only that the line's predictions, in standard units, lie nearer the mean than the values they are predicted from. Groups follow those predictions on average when the mean of one height is roughly a straight line in the other, as it is in these families; even then each group's mean scatters around its prediction, so no particular extreme group is guaranteed to sit nearer the mean.

If tall fathers have sons who are only tallish, and short fathers sons who are only shortish, will each generation be more average than the last, until after a few generations nearly everyone is about the same height? That does not happen: the sons' SD, 2.6475 inches, is slightly larger than the fathers', 2.5466. This is the **regression paradox**. Its resolution is that a son's actual height is not his predicted height: it is the prediction plus a part that his father's height does not foresee, and that part adds spread back (Gelman et al., 2020). The fitted values carry the part of a son's height that his father's height predicts, and spread only $r\,s_y = 1.1303$ inches; the residuals carry the rest, with spread $s_y\sqrt{1-r^2} = 2.3941$ inches (both SDs with the $n-1$ divisor; `sigma()` divides by $n-2$ and gives 2.4008). Variances add, not SDs: $1.1303^2 + 2.3941^2 = 2.6475^2$, which is Section 2.6's decomposition divided by $n-1$, in which the fitted part holds the share $R^2 = r^2 = .1823$. For this simple least-squares line, the predicted standard-unit distances shrink by the factor r, while the residual part keeps the outcomes more varied than their predictions. In a stable population, one that keeps its spread from generation to generation, the two parts balance (Gelman et al., 2020). Regression toward the mean describes predictions for selected groups, not a population growing more uniform.

The two spreads can be compared directly. Fit the sons' heights to the fathers' heights on the same records, then compare predictions with observed heights:

```
galton_fit <- lm(son ~ father, data = galton)
sd(fitted(galton_fit))
sd(galton$son)
```

7.3 Version two: one measurement taken twice

The second version measures one quantity twice, and an invented case shows a mechanism behind it. Imagine a large class of 2,000 students who sit a midterm and a final, with scores generated so that nothing between the exams has any effect. Each student has a stable true score T_i, and each exam adds luck independent of the true score and of the other exam's luck, with the same error SD on both exams. This is the true-score model of classical test theory that Part Two, Chapter 1, develops:

$$M_i = T_i + \varepsilon_{i1}, \qquad F_i = T_i + \varepsilon_{i2}, \qquad T_i \sim \mathcal{N}(65,\, 8^2), \qquad \varepsilon_{i1},\, \varepsilon_{i2} \overset{\text{ind}}{\sim} \mathcal{N}(0,\, 6^2).$$

Chapter 1 described reliability as the consistency of scores; here it is the share of score variance that comes from the true score, $8^2/(8^2 + 6^2) = .64$ on both exams. Only the true score repeats, so .64 is also the population correlation of the two exams, the slope of the line in standard units that the sample r estimates. The class is large so that its group means carry little chance variation. The mathematics is that of the heights, with the midterm in the father's place and the final in the son's. Four lines of R generate the scores:

```
set.seed(12345)
true_score <- rnorm(2000, mean = 65, sd = 8)
midterm <- true_score + rnorm(2000, mean = 0, sd = 6)
final <- true_score + rnorm(2000, mean = 0, sd = 6)
```

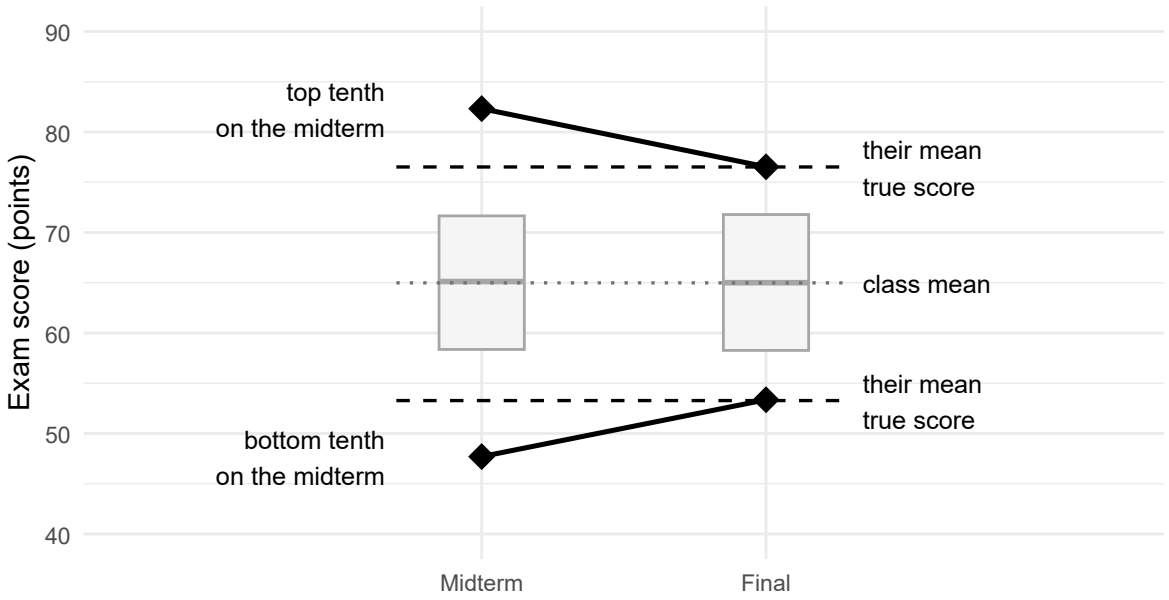


Figure 13: A hypothetical class of 2,000 students, each score a stable true score plus independent luck. The top and bottom tenth on the midterm (diamonds) score nearer the class mean on the final, near their unchanged mean true score (dashed); the boxes, the middle half of the class, span about the same range of scores on the final as on the midterm.

The simulated exams correlate at $r = .6509$, close to the population .64. In Figure 13 the 200 students in the top tenth on the midterm averaged 82.3217 points, and 76.5335 on the final: still well above the class mean, but nearer it. Their true scores average 76.5186, so the midterm overstated them. The least-squares line that predicts the final from the midterm has slope 0.667, below 1, which is regression toward the mean in points: a student 10 points above the midterm mean is predicted to score 6.6704 points above the final mean. At the top tenth's midterm mean, 82.3217, the line predicts 76.4691, close to where the group lands, 76.5335, and to its true-score mean.

Students picked for a high score are mostly strong but, on average, also lucky that day; on the final their luck is drawn afresh, and the group lands near its true level. The bottom tenth mirror them, rising from 47.6984 to 53.3550 toward a mean true score of 53.2729. A causal story is tempting: the top students grew overconfident and eased off, while the bottom students, alarmed by the midterm, worked harder. Neither story can hold here, because the scores come from a rule we wrote in which nothing happens between the exams: no true score changed, and no one was praised, tutored or discouraged (Gelman et al., 2020). The boxes show the other half of the story: the class is as spread out on the final (SD 10.0853) as on the midterm (SD 9.8406), because fresh luck carries other students into the extremes.

The stipulated Normal model makes the no-intervention prediction explicit: $E(F \mid M) = 65 + .64(M-65)$. A high midterm score combines an above-average true score with favorable luck; the next error has mean zero, while the true-score advantage remains. This known relationship comes from the generating rule. With true-score variation held fixed, less independent measurement error raises the correlation between exams and reduces expected regression toward the mean. A correlation below one alone would not establish the no-intervention relationship for an actual intervention study.

7.4 The regression fallacy

Regression toward the mean becomes a fallacy when the movement it predicts is credited to a cause. Tversky and Kahneman (1974) recount a conversation with flight instructors: a trainee praised for an unusually smooth landing tended to land less well next time, and a trainee reprimanded for a rough landing tended to do better, so the instructors concluded that praise hurts and criticism helps. As the authors note, a teacher who misses regression toward the mean will overrate criticism and underrate praise.

A quantitative understanding of prediction shows how this pattern can arise without an effect of feedback. Suppose a landing reflects stable skill plus independent luck on that attempt, as in the hypothetical class. Landings that draw comment are selected for being extreme; a later attempt is expected to be less extreme even if no feedback is given (Gelman et al., 2020). Crediting the whole movement to feedback without an adequate comparison is the **regression fallacy**. Feedback could still have an effect: the before-and-after pattern alone does not separate it from regression toward the mean or other changes.

The fallacy compares what happened with a hidden benchmark: a second measurement equal to the first, a son exactly as many SDs from his mean as his father (the dashed line of Figure 12), a final score equal to the midterm, a next landing as good as the last. That comparison assumes perfect prediction. In the programmed exam example, the known no-intervention mean is $65 + .64(M - 65)$, and its fitted line estimates that relationship. For an actual intervention, the benchmark must be the expected outcome **without the intervention**, supported by a suitable design or credible untreated data. An arbitrary fitted line, especially one fitted to outcomes already affected by feedback, cannot supply that benchmark by itself. Exercise 5 states an earlier-year no-tutoring relationship and asks what can be inferred if it also holds for this year's pupils.

The same mechanism can matter when cases are selected for an extreme first measurement and measured again: athletes and teams after a record season, pupils chosen for low test scores and given extra help, patients enrolled when their symptoms peak. Such groups can move toward the mean even without an intervention under a stable-score model with independent occasion errors. Random assignment of help among similarly selected cases makes both groups subject to the same selection mechanism, so their mean difference estimates the help's effect under the design's conditions. A nonrandom comparison selected and measured the same way also needs

a justification that it represents what would have happened without help (Barnett et al., 2005). Chapter 14, Section 4.1, applies this reasoning to selection on a noisy baseline.

8 Reading Bridge

A simple regression starts from a model: the mean of an outcome is a straight line in one predictor, with an intercept, a slope and an error SD, and a process whose values we set shows estimates landing near them, usually within about one SE. Applied to records, the line starts with a unit and a comparison. Its fitted values are the line's predictions for the records and its residuals the records' distances from them; least squares puts the line through the mean point, with the slope set by covariance and predictor variation. Across the 102 recorded occupations, fitted income rises by 0.8988 thousand 1971 Canadian dollars, about 899 dollars, per year of occupation-average education over the observed 6.38–15.97 years: a bounded description of the records, not an individual, causal or population return. Declaring the model for the occupations gives the slope a standard error and an interval, whose coverage repeated samples make visible. The residual SD, R^2, the slope's standard error, and the mean-response and new-case intervals answer different questions, and the checks say which answers the records support. That is the chapter's through-line: the line predicts, and, once a model is declared, standard errors and intervals say how uncertain its slope and predictions are. In standard units the line predicts r times a case's distance from the mean, whether the two measurements are a father's and a son's heights or one student's two exam scores. Groups selected for extreme values are predicted nearer the mean under the stated linear relation, while outcomes retain residual variation. Reading a before-and-after gain as an effect needs a credible no-intervention comparison and an appropriate causal design. Argument Chapter A reads a short report of the occupation line as an argument.

Chapter 8 writes the model with several predictors and reads a partial slope, first on a simulated process with two predictors. On the occupations it refits the education-only line on the 98 occupations with a recorded type, so that the change from 102 to 98 records is not mistaken for an effect of adjustment; it then adds prestige, percentage of women and occupation type, and asks a separate held-out prediction question. Chapter 9 takes Chapter 8's model as its map, changing its mean part with factors of several levels, model comparison and changed forms, and recomputing its uncertainty with robust and resampled methods; its Section 7 (Project STAR) reads a randomized class-size experiment whose uncertainty follows its school-clustered design (Finn & Achilles, 1990; Mosteller, 1995). Chapter 10 extends the factorial interactions of Chapters 4 and 6 to regression predictors.

Key terms

Use this short reference to revisit the regression concepts introduced here. The Part One Concept Reference gives the broader reference and the principal home of each concept; fitted values, residuals, standard errors, intervals, standardized values, and the reading vocabulary are explained in Chapters 1–4.

Table 11
Selected concepts for simple linear regression.

Term and common alternatives	Meaning
Simple linear regression (simple regression model; regression line)	A model in which an outcome's mean is a straight line in one predictor, $E(Y \mid x) = \beta_0 + \beta_1 x$, with errors around the line that average zero and share one SD σ; its three parameters are the intercept, the slope and σ. The least-squares line estimates the model's line: its descriptive use concerns the records fitted, and a model-based use needs a declared process.
Intercept	The fitted outcome at predictor value zero. It places the line vertically; when zero is outside the observed support, it is a mathematical anchor rather than a supported prediction. Centering the predictor moves it to a supported reference point.
Slope (regression coefficient)	The difference in fitted outcome per one-unit difference in the predictor, in outcome units per predictor unit. It compares units that differ in the predictor; it is not a change within one unit unless a design supports that reading.
Ordinary least squares (OLS)	The rule that chooses the intercept and slope minimizing the residual sum of squares. With an intercept, the fitted line passes through the mean point and the residuals sum to zero; the slope equals covariance divided by predictor variance.
Residual sum of squares (RSS)	The sum of squared residuals from a fitted line, in squared outcome units. Total squared variation about the mean splits into a regression part and RSS when the model has an intercept.
Residual standard deviation (residual standard error)	The square root of RSS divided by its residual degrees of freedom, in outcome units; with an intercept and one predictor the degrees of freedom are $n - 2$. Under the process model it estimates the error SD.
R-squared (coefficient of determination)	One minus RSS divided by total squared variation: the proportion of centered squared outcome variation accounted for by the fit in these records. In simple regression with an intercept it equals the squared correlation.

Term and common alternatives	Meaning
Confidence interval for a mean response (mean-response interval)	An interval for the process mean outcome at a stated predictor value, using the SE of the fitted value. It concerns an average, narrows near the predictor mean, and does not cover individual outcomes.
Prediction interval (new-case interval)	An interval for one new outcome at a stated predictor value from the same process. Its variance adds the error variance to the fitted-mean variance, so it is wider than the mean-response interval and keeps a nonzero floor as the sample grows.
Leverage	A measure of how unusual a record's predictor value is relative to the others; it grows with distance from the predictor mean and appears in the SE of a fitted value and in influence measures.
Cook's distance (influence)	A summary of how much the fitted values change when one record is omitted, combining its residual and its leverage. It prompts inspection of provenance and a disclosed sensitivity, not automatic deletion.
Heteroskedasticity-consistent covariance (HC3; sandwich estimator)	A coefficient covariance estimate that allows unequal error variance across predictor values under independence. It changes SEs and intervals, not the fitted coefficients, and does not repair curvature, dependence, measurement error, or confounding.
Extrapolation (predictor support)	Using a fitted line at predictor values outside the observed range or where observations are sparse. Software returns a number there; the data do not support it.
Regression toward the mean (regression to the mean)	For a simple linear prediction in standard units, $\hat{z}_y = r\, z_x$; when $\lvert r \rvert < 1$, the predicted value is nearer its mean than the predictor is to its mean. In a stable repeated-measurement model with independent occasion errors, a group selected on an extreme noisy first score is expected to be less extreme on the second. These are statements about predictions or conditional averages, not every case; a narrower spread of predictions does not imply a narrower outcome distribution (the regression paradox).
Regression fallacy	Attributing to an intervention the movement expected from selection on an extreme noisy measurement. Evaluating an intervention requires a credible no-intervention benchmark, supplied by a suitable design or untreated comparison under stated assumptions; a fitted line through treated outcomes alone does not supply it.

Further reading

For another accessible account of lines, residuals, and fit, see OpenIntro's single-predictor regression tutorial. For a compact executable route from a fitted model to intervals and checks, see the ISLR authors' Chapter 3 R lab (James et al., 2021). Consult the carData reference manual (Fox et al., 2022) for data documentation, Barnett et al. (2005) on regression to the mean in repeated measurements, and Fox (2016), Fox & Weisberg (2019), and Gelman et al. (2020) for broader regression treatments.

Exercises

The prompts below use **Foundational** and **Integrative** levels. Where rounded summaries are supplied, answers within 0.02 in the stated units are sufficient. Exercises 1–6 draw on the main chapter.

Exercise 1: Read the unit and the line (Foundational)

A teaching report describes 80 occupations with $\widehat{\text{income}} = -1.5 + 0.7\,\text{education}$, where income is in thousand dollars and average education ranges from 7 to 16 years. Interpret the slope and the intercept. Calculate the fitted income at 12 years and the residual for an occupation at 12 years with income 9. Explain why the slope does not give an individual's gain from another year of schooling.

Exercise 2: Connect covariance and the fitted line (Foundational)

A set of records has $\bar{x} = 4, \bar{y} = 10, S_{xx} = 20, S_{xy} = 30$. Calculate the OLS slope and intercept, and the fitted value at $x = 5$. Show that the line passes through the mean point. If $s_x = 2$ and $s_y = 5$, calculate the correlation implied by these summaries. What would prevent fitting a slope if every x were equal?

Exercise 3: Separate fit and uncertainty (Foundational)

An intercept-and-slope regression has $n = 22$, $RSS = 80$, $TSS = 144$, $S_{xx} = 100$, and fitted slope 0.8. Calculate residual SD, R^2, and the classical slope SE. Use $t_{.975,20} = 2.086$ to obtain a 95% slope interval. State the process assumptions needed for that interval and explain why it does not by itself establish that the line fits closely or identifies a cause.

Exercise 4: Mean or new case? (Foundational)

Under the stated independent Normal-error constant-variance model, a fitted value at $x_0 = \bar{x}$ is 12, $n = 25$, and $s_e = 3$. Before calculating, consider a larger study with 100 independent records, the same predictor mean and the same residual SD. Predict whether the mean-response SE and the new-case prediction SE would both halve at $x_0 = \bar{x}$, and explain your prediction. Then, for the original 25-record study, use $t_{.975,23} = 2.069$ to calculate the SE and 95% interval for the conditional mean and the prediction SE and interval for one independent new case at the same x_0. Explain the extra 1 in the prediction expression. Would either interval validate the model in a different population?

Exercise 5: Tell regression toward the mean from an effect (Integrative)

In an invented study, a school records a reading test in September and again in June for all 200 of its pupils. Both tests have mean 50 and SD 10, and their correlation is .6. The 20 pupils with the lowest September scores, averaging 32 points, receive extra tutoring, and their June mean is 42.8 points. These figures include the tutored pupils' June scores, so by themselves they cannot show what those scores would have been without tutoring. The school's records from earlier years, when no pupil was tutored, supply that benchmark: the two tests again had means 50, SDs 10 and correlation .6, and the mean June score of pupils with the same September score followed a straight line in that score. Assume that this year's pupils would have followed the same pattern without tutoring. (1) Express the tutored group's September mean in standard units, and give the June mean that this pattern's line in standard units predicts for pupils with that September score, in standard units and in points. (2) The school reports that tutoring raised the group by 10.8 points. Name the error in reading the whole gain as an effect of tutoring, and say how much of the gain regression toward the mean predicts by itself under this pattern. (3) The report adds that, since the lowest scorers caught up, the June scores of all 200 pupils must be less spread out than September's. Split a June score into the part that the September score predicts and the part it does not, give the SD of each part, and explain why regression toward the mean leaves the June SD at 10. (4) Describe a comparison that would separate an effect of tutoring from regression toward the mean.

Exercise 6: Respond to sensitivity (Integrative)

In a descriptive analysis of 102 occupations, the slope is 0.9. A valid, high-income occupation has the largest Cook's distance; removing it gives 0.79. A curved-fit check changes the shape, and an HC3 calculation changes the slope SE from 0.13 to 0.15 without changing the line.

What does each check tell you? What does it not tell you? Explain why deleting the occupation automatically is inappropriate. Distinguish a variance repair for the same estimand from replacing a constant process slope with a different estimand. Write a short sensitivity disclosure for the descriptive analysis.

Appendix A: Extended regression mathematics and interval derivations

This appendix derives the identities the main text uses, keeping identities about the observed records apart from claims about hypothetical repetitions. Occupation calculations use the same 102 rows, education in years and income in thousand 1971 Canadian dollars.

Derivatives, normal equations, and the minimizing line

Start with a fitted model that predicts one common center m for every observation. Chapter 2 Section 2.2 states that the mean minimizes the sum of squared deviations, and Chapter 3 Appendix A derives the identity behind it:

$$L(m) = \sum_i (y_i - m)^2 = \sum_i (y_i - \bar{y})^2 + n(m - \bar{y})^2.$$

Write $y_i - m = (y_i - \bar{y}) + (\bar{y} - m)$ and expand the square. The cross term is zero because $\sum_i (y_i - \bar{y}) = 0$. For $n > 0$, the last term is nonnegative and vanishes only at $m = \bar{y}$, so the mean uniquely minimizes this criterion.

The three outcomes $2, 4, 3$ from Section 2.4 have mean 3 and $L(m) = 2 + 3(m - 3)^2$, so their best constant line has loss 2. Letting the fitted value depend on the three predictor values reduces RSS to 1.5, as Table 4 showed; neither result requires Normal errors or a probability sample. Chapter 6's likelihood example chose a parameter by optimizing a different criterion; here we develop the least-squares line algebraically.

The intercept-and-slope criterion has two coefficients to determine, so we set its two partial derivatives to zero and then verify that the resulting line is the minimum. For candidate intercept b_0 and slope b_1, define

$$Q(b_0, b_1) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2.$$

Differentiating with respect to each candidate coefficient gives

$$\frac{\partial Q}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \qquad \frac{\partial Q}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

At the minimum both derivatives are zero. The resulting **normal equations** are

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i = \sum_i y_i, \qquad \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2 = \sum_i x_i y_i.$$

The first equation gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Substitution into the second yields

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = \frac{S_{xy}}{S_{xx}}.$$

Here $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$ is required: a slope cannot be identified when every predictor value is the same. Because sample covariance and variance use the same denominator $n - 1$,

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

In the occupation data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} \approx 0.8988, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx -2.8536.$$

Unlike a correlation, the slope changes when a variable's units change. Standardizing both variables makes their SDs one, so the slope becomes r; Section 7 reads that line.

To verify that the stationary point is the unique minimum, write any other line as $\hat{\beta}_0 + d_0 + (\hat{\beta}_1 + d_1)x_i$. The normal equations make the cross term with the OLS residuals vanish, leaving

$$Q(\hat{\beta}_0 + d_0, \hat{\beta}_1 + d_1) = RSS + n(d_0 + d_1\bar{x})^2 + d_1^2 S_{xx}.$$

With $S_{xx} > 0$, every different line has larger RSS. If a candidate line is constrained to pass through $(\bar{x}, \bar{y})$, its excess RSS is simply $S_{xx}(b_1 - \hat{\beta}_1)^2$. Section 2.5 makes this comparison visible with five candidate lines.

Residual constraints and the decomposition of variation

At the OLS solution the derivative equations become $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Thus the residuals have zero sample covariance with the predictor and mean zero **by construction**. They are not evidence that the unknown population errors are independent, normally distributed, or conditionally mean zero.

The fitted line passes through $(\bar{x}, \bar{y})$, so $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$. Its centred fitted values therefore have zero cross-product with the residuals:

$$\sum_i (\hat{y}_i - \bar{y})e_i = \hat{\beta}_1 \left(\sum_i x_i e_i - \bar{x} \sum_i e_i \right) = 0.$$

Writing each observed deviation as its fitted part plus its residual, then expanding the square, gives

$$\begin{aligned}\sum_i (y_i - \bar{y})^2 &= \sum_i \{(\hat{y}_i - \bar{y}) + e_i\}^2 \\ &= \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i e_i^2,\end{aligned}$$

since $\sum_i (\hat{y}_i - \bar{y}) e_i = 0$. This proves $TSS = RegSS + RSS$, an algebraic decomposition of observed variation, not a division into causal contributions. For these records, $1820.8134 = 607.4214 + 1213.3920$, in squared thousand-dollar units.

Also $RegSS = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$. Dividing by $TSS = S_{yy}$ proves $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$ for simple regression **with an intercept**. The identities need not hold for a no-intercept fit. When fitted values are nonconstant, $\text{cor}(y, \hat{y})^2 = R^2$ as well.

The two fitted coefficients impose the two residual constraints. Under a correct linear mean with independent errors of common variance, the same least-squares identity applied to the errors gives

$$RSS = \sum_i \varepsilon_i^2 - n\bar{\varepsilon}^2 - \frac{\{\sum_i (x_i - \bar{x})\varepsilon_i\}^2}{S_{xx}}.$$

The expectations of the three terms are $n\sigma^2$, σ^2 and σ^2. Thus $E(RSS) = (n-2)\sigma^2$, and $s_e^2 = RSS/(n-2)$ is an unbiased estimate of the error variance. Its square root s_e estimates the error SD in the outcome's units but is not itself exactly unbiased for σ.

From an error model to slope and intercept uncertainty

The 102-record least-squares line is already determined by those records. The next derivations concern a **separate model-based question**: the coefficients of a process that could generate new, independent occupation outcomes at the observed education values. They assume the model of Section 1.1 at fixed observed $x_1, \ldots, x_n$, with $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. These are working conditions, not consequences of fitting `lm()`. Section 3.1 derives the slope's variance, σ^2/S_{xx}, and its SE, $s_e/\sqrt{S_{xx}}$. Because $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and $\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0$,

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \qquad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Here $S_{xx} = \sum_i (x_i - \bar{x})^2$, as in Section 2.4. The covariance is why simply adding the variances of an uncentered intercept and slope would be wrong; with the predictor centered at its mean the two estimates are uncorrelated, and the intercept SE is $s_e/\sqrt{n}$.

Under the stated linear-mean, uncorrelated, equal-variance conditions, OLS is the best linear unbiased estimator: it has the smallest variance among estimators linear in the outcomes and unbiased for the coefficients. This Gauss–Markov statement does not require normal errors and does not establish causal identification, valid measurement, or population coverage (Fox, 2016; Kutner et al., 2005).

If the independent errors are also Normal, then the standardized coefficient error is normal, RSS/σ^2 has a χ^2_{n-2} distribution, and these quantities are independent. Their ratio gives

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-2}, \qquad j = 0, 1.$$

That is the basis of exact classical coverage for $\hat{\beta}_j \pm t_{.975,n-2} SE(\hat{\beta}_j)$ and of the corresponding null test, which Section 3.1 applies to the occupation slope; the diagnostics of Section 6 do not support treating that interval's nominal coverage as established.

These derivations condition on the predictor values, which need not have been fixed by an experiment; a model with random predictors uses the same conditional argument when its error conditions hold. Neither route shows that the occupations were probability sampled, and neither requires education or income to be Normal.

Why a new-case interval has an additional variance term

For a supported value x_0, define the model mean $\mu_0 = \beta_0 + \beta_1 x_0$ and its estimate $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Substituting the coefficient variances and covariance above gives

$$\begin{aligned} \operatorname{Var}(\hat{\mu}_0) &= \operatorname{Var}(\hat{\beta}_0) + x_0^2 \operatorname{Var}(\hat{\beta}_1) + 2x_0 \operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left\{ \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}. \end{aligned}$$

All predictor values, including the chosen x_0, remain fixed. A confidence interval for this mean uses the square root with s_e for σ. A new occupation outcome is instead $Y_0 = \mu_0 + \varepsilon_0$; if its error is independent of the fitted data and comes from the same Normal, equal-variance process,

$$\operatorname{Var}(Y_0 - \hat{\mu}_0) = \sigma^2 \left\{ 1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}.$$

The extra 1 is new outcome variation, which would remain even if the model mean were known. Both intervals use the same $t_{.975,n-2}$ multiplier. The table applies both formulas at four supported education values, the same calculation as Section 5's two `predict()` calls.

Table 12
Mean-response and new-case intervals at four supported education values.

Education	Fitted	Mean lower	Mean upper	Case lower	Case upper
8	4.3369	3.3651	5.3087	−2.6420	11.3158
10	6.1345	5.4254	6.8437	−0.8127	13.0818
12	7.9322	7.1776	8.6868	0.9802	14.8842
15	10.6286	9.3550	11.9022	3.6013	17.6559

At 10 years the point prediction is 6.1345 thousand dollars, the mean interval [5.4254, 6.8437] and the new-occupation interval [−0.8127, 13.0818]. Cutting the negative endpoint off at zero would not produce a valid 95% interval; a transformation or a model that respects the boundary would be a different analysis with its own checks.

None of these formulas includes uncertainty from a wrong mean function, biased measurement or a shift to another labour market. They are pointwise model intervals, not a simultaneous band or an out-of-sample validation, and their target is a new occupation-level record from the declared process, never an individual worker's income.

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